Protocol

Work protocol for the preparation of electricity consumption datasets for theft detection

Version 1.1 | Collection CER4 | Issued 27 October 2026

Abstract. This protocol records the work carried out from the start of the research to the preparation of the collection CER4: the inputs provided, the acquisition and verification of four public datasets, their cleaning, their reduction to one common daily record, the normalization, the final window form, the checks applied and the delivery as Excel workbooks. Every entry carries the date and time at which it was recorded. All times are UTC. The revision history states what version 1.1 changed. The protocol ends with the procedure for converting the Markdown and LaTeX versions of the documents to PDF and other formats.

Keywords: work protocol; electricity theft detection; smart meter data; data provenance; reproducibility

1 Document Control

Table 1. Document control.

Item	Entry
Title	Protocol, work protocol for the preparation of electricity consumption datasets for theft detection
Project	Cross Region Generalisation in Electricity Theft Detection
Version	1.1, which revises version 1.0 of collection CER3
Collection	CER4
Issue date	27 October 2026, 06:33 UTC
Time basis	UTC. Local time of the work computer is UTC plus 5 hours. A time marked not recorded was not stored by the source.
Scope	SGCC, CER, London LCL and ETD2022: acquisition, verification, cleaning, common daily record, normalization, final window form and delivery as Excel workbooks
Related documents	Preprocessing Protocol, Normalization Protocol and Summary Report of the same collection
Formats	Every document in PDF, Markdown and LaTeX. Datasets as Excel workbooks (xlsx)

Table 2. Revision history.

Version	Date (UTC)	Collection	Changes
1.0	27 October 2026, 04:49	CER3	First issue. The datasets as 38 Excel workbooks. The four documents in PDF, Markdown and LaTeX.
1.1	27 October 2026, 06:33	CER4	Fifteen workbooks that held only a selection of the rows of a table were removed. Eleven workbooks that repeat the content of other workbooks or are not needed to use the data were removed: the window indexes, the run audit, the household and block attribute tables, and the ETD2022 hourly tables. The final window tables are delivered complete, in two workbooks for each of the five seeds. The SGCC daily table is delivered complete, in 44 workbooks. The documents were revised to match.

2 Inputs Provided

Table 3 lists the documents and files that were provided at the start of the work, with the date and time recorded for each. The documents I1 and I2 carry the time of their creation. The other entries carry the time of the file or of the log entry.

Table 3. Register of inputs.

Ref	Input	Content	Recorded (UTC)	Used for
I1	Research Protocol, 4 pages	Problem statement, research questions RQ1 to RQ5, hypotheses H1 to H5, class definition, metrics and success criteria	10 August 2026, 19:42	Positive class is theft. Evaluation design with five seeds
I2	Research Plan, 4 pages	Workflow S0 to S7, dataset roles, preprocessing steps, normalization views, data splits, models	11 August 2026, 05:21	Daily resolution, interpolation, absolute and relative scales, split ratios, leakage control
I3	Systematic review, 9 pages	Guided systematic review of 188 records on electricity theft detection, 2017 to 2026	3 September 2026, date in the file name, time not recorded	Background and context of the datasets
I4	CER download script	List of the files of the CER package	8 September 2026, 13:55	Acquisition of CER
I5	CER trial package	Six archives of half hourly readings, trial manifest, allocation file, survey files, findings report	8 September 2026, 14:04 to 15:07	Target dataset CER
I6	SGCC package	Daily readings of 42,372 consumers: data.csv, two archive parts and a README	19 September 2026, 19:27	Source dataset SGCC
I7	Cleaning and normalization instructions	Gap, outlier, negative value and scale rules given by the supervisor	Time not recorded. In force from the first preprocessing run, 20 October 2026, 19:21	Cleaning rules and their precedence over the research plan
I8	Source verification specification	Seven numbered instructions for the verification of the LCL, ETD2022, SGCC and CER sources	21 October 2026, 16:34	Verification of sources

3 Work Log

Table 4 lists the work in order of the recorded time. Counts are those of the delivered tables.

Table 4. Work log.

No	Date	Time (UTC)	Stage	Action	Result
W01	10 August 2026	19:42	Inputs	Research Protocol created (I1)	Research questions, hypotheses, metrics
W02	11 August 2026	05:21	Inputs	Research Plan created (I2)	Workflow, dataset roles, preprocessing, splits
W03	3 September 2026	Not recorded	Inputs	Systematic review dated (I3)	Background for the dataset selection
W04	8 September 2026	13:55	Acquisition	CER download script saved (I4)	Lists the files of the CER package
W05	8 September 2026	14:04 to 15:07	Acquisition	CER package saved (I5): six reading archives, File1 at 14:04 and File6 at 15:02, then manifest, allocation, survey and report files	Package complete
W06	19 September 2026	19:27	Acquisition	SGCC package saved (I6): data.csv, two archive parts, README	data.csv of 175,194,613 bytes
W07	20 September 2026	06:54 to 06:56	Verification	SHA256 and ZIP integrity of the six CER archives	All six passed
W08	20 September 2026	06:56 and 07:01	Verification	Content analysis of the CER archives	157,992,996 raw rows, 6,435 meters

Table 4 (continued). Work log.

No	Date	Time (UTC)	Stage	Action	Result
W09	20 September 2026	07:02	Verification	SHA256 of the CER allocation file recorded	Recorded in the manifest
W10	20 September 2026	07:03	Acquisition	SGCC release downloaded from the GitHub repository, three files	52,470,873 bytes
W11	20 September 2026	07:07	Verification	SGCC split archive extracted	data.csv of 175,194,613 bytes, SHA256 beginning 99f8fd31
W12	20 September 2026	07:09	Verification	Kaggle copy of SGCC hashed	SHA256 identical to the GitHub copy
W13	20 September 2026	07:13	Verification	Local SGCC files hashed	Identical to the GitHub and Kaggle copies
W14	20 September 2026	07:20	Verification	LCL Kaggle package hashed	1,251,594,914 bytes, SHA256 beginning 8839527a
W15	20 September 2026	07:21	Acquisition	114 files extracted from the LCL package: half hourly blocks, household table, ACORN table	CRC passed for every file
W16	20 September 2026	08:07	Acquisition	LCL tariff workbook downloaded	245,384 bytes
W17	20 October 2026	18:56 to 18:57	Inputs	Research documents I1 to I3 saved on the work computer	Files of 253,820, 352,455 and 901,853 bytes
W18	20 October 2026	19:21	Preprocessing	SGCC daily table built	43,855,020 rows, 42,372 consumers, 3,615 labelled theft
W19	20 October 2026	19:23 to 19:26	Preprocessing	LCL household table and ACORN category table built	5,566 households, 826 category rows
W20	20 October 2026	19:40	Preprocessing	LCL half hourly and daily tables built	168,268,119 and 3,507,915 rows
W21	20 October 2026	19:43 to 19:44	Preprocessing	CER half hourly and daily tables built	158,484,746 and 3,301,514 rows, recorded complete
W22	20 October 2026	20:32	Acquisition	LCL official archive obtained from the London Datastore	801,674,949 bytes, SHA256 beginning 68a35598
W23	21 October 2026	02:37	Verification	Official LCL archive tested, and its household streams compared with the Kaggle package	Integrity and publisher ETag passed. 5,566 streams equal after removal of 115,453 exact duplicate rows
W24	21 October 2026	16:31	Records	Source manifest written	File, source, version, licence, size and SHA256 of every accepted file
W25	21 October 2026	16:34	Acquisition	ETD2022 version 3 downloaded from Mendeley Data (verification specification I8 dated the same minute)	64,403,651 bytes, size and SHA256 beginning b10ea864 match the publisher
W26	23 October 2026	17:29	Preprocessing	ETD2022 audited and cleaned	560,655 rows, 15 quarantined, 560,640 clean, 16 archetypes, 64 blocks, 23,360 relative days
W27	23 October 2026	17:28 to 17:31	Records	Audit observations and audit database written	Counts of every dataset indexed
W28	25 October 2026	14:55	Preprocessing	ETD2022 daily table built on the common basis	23,360 rows, nominal calendar from 1 January 1971, daily kWh recomputed from the hourly table

Table 4 (continued). Work log.

No	Date	Time (UTC)	Stage	Action	Result
W29	25 October 2026	14:56	Preprocessing	Common daily table of the four datasets built	50,687,809 rows
W30	25 October 2026	14:57	Normalization	Partitions of five seeds assigned, scaler parameters fitted on SGCC training consumers, meter means and window index built	272,160 assignments, 1,275,674 windows, 108,930 contiguous runs
W31	25 October 2026	14:57	Normalization	Normalized daily tables built for five seeds, 14:57:29 to 14:57:50	39,712,070 rows each
W32	25 October 2026	15:01	Final form	Final window tables built for five seeds, 15:01:29 to 15:01:45	1,275,674 windows each
W33	25 October 2026	15:03	Verification	Independent validation of stages 1 to 3	41 checks passed
W34	27 October 2026	04:29	Delivery	Export of Excel workbooks for collection CER3 started	Whole tables, parts of tables and extracts of tables
W35	27 October 2026	04:35	Delivery	Export for collection CER3 finished	38 workbooks, 0.73 GB
W36	27 October 2026	04:38 to 04:45	Verification	Workbooks of collection CER3 compared with their source tables	38 of 38 workbooks verified
W37	27 October 2026	04:49	Documents	Version 1.0 of the four documents issued in PDF, Markdown and LaTeX	Collection CER3
W38	27 October 2026	05:11	Delivery	Export of Excel workbooks for collection CER4 started: final window tables of five seeds, stage 2 tables, stage 1 tables of CER, London LCL and ETD2022	22 workbooks planned
W39	27 October 2026	05:37	Delivery	Export of the first group finished	22 workbooks, 4.62 GB
W40	27 October 2026	05:37	Delivery	Export of the SGCC daily table started	44 workbooks planned
W41	27 October 2026	06:01	Delivery	Export of the SGCC daily table finished	44 workbooks, 1.84 GB
W42	27 October 2026	05:38 to 06:32	Verification	Workbooks of collection CER4 compared with their source tables	66 of 66 workbooks verified
W43	27 October 2026	06:33	Documents	Version 1.1 of the Protocol, Preprocessing Protocol, Normalization Protocol and Summary Report issued in PDF, Markdown and LaTeX	Collection CER4

4 Modification Register

Table 5 lists every change made to the data. Raw files were not modified. Each modification is a rule applied to a derived table and is counted in the Summary Report.

Table 5. Modifications to the data.

Ref	Dataset	Modification	Count
M01	All	Raw source files left unchanged	0 files modified
M02	SGCC	Interior gaps of at most three days filled by linear interpolation	326,500 days
M03	SGCC	Date 18 September 2016, absent in the source, interpolated as a one day gap	39,710 consumers
M04	SGCC	Long gaps and boundary gaps left empty, days marked ineligible	10,949,400 days
M05	SGCC	Readings above five times the meter's 99th percentile flagged, not changed	872 readings
M06	SGCC	Consumers without any valid reading marked ineligible	5 consumers
M07	CER	Conflicting duplicate slots invalidated and their days quarantined	12 slots
M08	CER	Rows in slots that cannot occur on the local day removed	560 rows
M09	CER	Interior gaps of at most three intervals interpolated	12 intervals
M10	CER	Long gaps and boundary gaps left empty	492,322 intervals
M11	CER	Readings above five times the 99th percentile flagged	682 readings
M12	CER	Slots converted to Dublin civil days with 46, 48 or 50 slots	3,301,514 days
M13	London LCL	Exact duplicate rows of the publisher file removed	115,453 rows
M14	London LCL	Off grid null rows removed	5,560 rows
M15	London LCL	Interior gaps of at most three intervals interpolated	19,224 intervals
M16	London LCL	Long gaps left empty	437,434 intervals
M17	London LCL	Readings above five times the 99th percentile flagged	1,269 readings
M18	London LCL	Interval end stamps shifted back by 30 minutes to form GMT days	3,507,915 days
M19	ETD2022	Rows in which every feature is zero quarantined	15 rows
M20	ETD2022	Hourly kW summed to daily kWh, each hour multiplied by one hour	23,360 days
M21	ETD2022	64 blocks of 8,760 rows named as meters and given a nominal calendar from 1 January 1971	64 blocks
M22	ETD2022	Day label set to theft when any hour carries an attack label	10,221 days
M23	All	Common daily record with the column calendar_basis added	50,687,809 rows
M24	All	Columns split, kwh_absolute_minmax and kwh_relative_meter_mean added in five seeds	39,712,070 rows per seed
M25	All	Windows of 30 consecutive eligible days formed. Remaining days unused	1,275,674 windows, 1,441,850 days
M26	Delivery	Tables larger than one Excel sheet delivered in consecutive numbered workbooks of at most 1,000,000 rows	66 workbooks

5 Deliverables

All paths are relative to the collection folder CER4. An Excel sheet holds at most 1,048,576 rows. A table that fits one sheet is delivered in one workbook. A larger table is delivered in consecutive numbered workbooks of at most 1,000,000 rows, each with its own header, and the parts together hold every row of the table.

Table 6. Contents of the collection.

Folder	Content	Workbooks
`1_preprocessing`	Cleaned daily table of each dataset, in one subfolder for each of SGCC, CER, London LCL and ETD2022	53
`2_normalization`	Partitions of the meters for five seeds, scaler parameters of the five seeds, and the mean of each meter	3
`3_final_common_normalized`	One record of 30 daily values per window for all four datasets, two workbooks for each seed	10
`Documents/PDF, Markdown, LaTeX`	Protocol, Preprocessing Protocol, Normalization Protocol and Summary Report in three formats, with the figure of the Summary Report beside the Markdown and LaTeX files	14 files

6 Verification Record

Table 7. Checks and their outcome.

Check	Method	Outcome	Recorded (UTC)
Source integrity	SHA256, ZIP CRC, publisher ETag, per household stream comparison	All passed	20 September 2026, 06:54 to 21 October 2026, 16:34
Reconciliation of readings	Expected readings equal readings present plus missing, for each dataset	Exact	20 October 2026, 19:21 to 23 October 2026, 17:29
ETD2022 hourly to daily	Daily kWh recomputed from the hourly table, labels and nominal days compared	Equal, relative difference at most 0.00000006	25 October 2026, 15:03
Normalization	Partitions, scaler range 0 to 1, formulas on every value, window length and identifiers, window values against daily values	41 checks passed	25 October 2026, 15:03
Independent normalization run	Values, partitions, scalers and windows of SGCC, CER and London LCL compared with a run of 23 October 2026, 17:26	Identical	25 October 2026, 14:57 to 15:03
Workbooks	Row count, header, frozen header row, and per column count, sum, minimum, maximum, distinct count or count of true values against the source table. The parts of a table add up to its rows	66 of 66 verified	27 October 2026, 05:38 to 06:32

7 Standing Limits

The limits of the data are stated in the Preprocessing Protocol, Section 9, and in the Normalization Protocol, Section 8. They concern the differing calendar periods of the datasets, the compression of typical values by the literal minimum and maximum scale, the consumer level labels of SGCC, the absence of theft labels in CER and London LCL, and the inferred blocks and nominal calendar of ETD2022.

8 Converting the Documents

Every document is provided as PDF, Markdown and LaTeX in the folders `Documents/PDF`, `Documents/Markdown` and `Documents/LaTeX`. The PDF files need no conversion. The procedures below produce other formats from the Markdown and LaTeX files. Commands are written for a terminal opened in the folder that holds the file. Pandoc is the converter used in this section.

8.1 Markdown to Word, HTML and other formats

These conversions need Pandoc only. The Summary Report refers to the image `coverage_timeline.png`, which must stay in the same folder as `Summary_Report.md`.

Table 8. Pandoc commands for Markdown files.

Target format	Command
Word document	`pandoc Protocol.md -o Protocol.docx`
HTML page with styles in one file	`pandoc Protocol.md -s --embed-resources -o Protocol.html`
OpenDocument text	`pandoc Protocol.md -o Protocol.odt`
Plain text	`pandoc Protocol.md -t plain -o Protocol.txt`
LaTeX source	`pandoc Protocol.md -s -o Protocol_from_markdown.tex`

8.2 Markdown to PDF

Pandoc makes a PDF only through a LaTeX engine, which is not part of Pandoc. Install MiKTeX or TeX Live first, then run the command below. Without a LaTeX engine, use one of the two routes that need no installation: convert to a Word document as in Section 8.1, open it in Word and save it as PDF, or convert to an HTML page and print it to PDF from a browser.

Table 9. Routes from Markdown to PDF.

Route	Steps
Pandoc with LaTeX	`pandoc Protocol.md -o Protocol.pdf --pdf-engine=xelatex -V geometry:margin=2.5cm`
Word	Convert to a Word document, open it in Word, then File, Save As, PDF
Browser	Convert to an HTML page, open it in a browser, then Print, Save as PDF
Editor	Open the Markdown file in an editor that exports PDF, for example Visual Studio Code with the extension Markdown PDF, then export

8.3 LaTeX to PDF

The LaTeX files are complete documents that use only standard packages: `geometry`, `longtable`, `booktabs`, `array`, `fancyhdr`, `caption`, `graphicx`, `hyperref`, `mathptmx` and `courier`. Compile each file twice so that page references settle. The file `Summary_Report.tex` needs `coverage_timeline.png` in the same folder.

Table 10. Compiling the LaTeX files.

Route	Steps
Local installation	`pdflatex Protocol.tex`, run twice. `xelatex Protocol.tex` gives the same result
Online editor	Create a project in an online LaTeX editor such as Overleaf, upload the .tex files and the image, and compile with pdfLaTeX
All documents	`pdflatex Preprocessing_Protocol.tex`, `pdflatex Normalization_Protocol.tex` and `pdflatex Summary_Report.tex`, each run twice

Note. The LaTeX files were checked for structure by reading them back with Pandoc. They were not compiled on the computer used for preparation, because no LaTeX engine is installed there.

8.4 Other formats

A Word document from Section 8.1 can be saved as OpenDocument or PDF from Word. An HTML page can be opened in Word. The Excel workbooks open directly in Excel. A table of more than 1,000,000 rows is split into numbered parts, as stated in the Preprocessing Protocol, Section 10, and in the Normalization Protocol, Section 10.